

The logarithmic Brunn–Minkowski conjecture

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Abstract

We prove the logarithmic Brunn–Minkowski conjecture for arbitrary origin-symmetric convex bodies in every dimension. The theorem also gives the symmetric L_p Brunn–Minkowski inequality for every $0 < p < 1$. Combined with Saroglou’s transfer theorem and a support-subspace reduction, it yields the logarithmic inequality for every even log-concave Radon measure and the scalar-dilation (B) -conjecture.

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1 Introduction

A convex body in $\mathbb{R}^n$ is a compact convex set with nonempty interior. If $K = -K$, then 0 lies in its interior. Its support function is

$$h_K(u) = \max_{x \in K} \langle x, u \rangle, \quad u \in S^{n-1}.$$

For an origin-symmetric body this function is strictly positive. For a positive continuous function f on the unit sphere, its *Wulff body* is

$$\mathcal{W}[f] = \bigcap_{u \in S^{n-1}} \{x \in \mathbb{R}^n : \langle x, u \rangle \leq f(u)\}.$$

It is a convex body: it contains the ball of radius $\min f > 0$ and is contained in the ball of radius $\max f$. Write $|K|$ for n -dimensional Lebesgue measure. Our main result is the following.

Theorem 1.1 (Even logarithmic Brunn–Minkowski inequality). *Let $n \geq 1$ and let $K, L \subset \mathbb{R}^n$ be convex bodies satisfying $K = -K$ and $L = -L$. For every $0 \leq \lambda \leq 1$,*

$$|\mathcal{W}[h_K^{1-\lambda} h_L^\lambda]| \geq |K|^{1-\lambda} |L|^\lambda. \quad (1.1)$$

The interpolation in (1.1) takes the geometric mean of the support bounds before intersecting the half-spaces. This gives a strengthening, in the symmetric setting, of the multiplicative form of the classical Brunn–Minkowski inequality: the arithmetic–geometric mean inequality gives

$$\mathcal{W}[h_K^{1-\lambda} h_L^\lambda] \subset (1-\lambda)K + \lambda L.$$

In general $h_K^{1-\lambda} h_L^\lambda$ need not itself be a support function. The intersection defining $\mathcal{W}$ is essential throughout the argument.

For $0 < p < 1$, the L_p interpolation replaces the geometric mean of the support bounds by the weighted power mean $((1-\lambda)h_K^p + \lambda h_L^p)^{1/p}$, still taking the Wulff body. The logarithmic endpoint implies the full symmetric volume inequality in this intermediate range. Böröczky, Lutwak, Yang and Zhang record this monotone implication [6, equation (1.10) and the following paragraph, p. 1977]. The additive volume–power form below follows by normalizing the bodies and reweighting the interpolation parameter.

Corollary 1.2 (Symmetric L_p Brunn–Minkowski inequality). *Let $n \geq 1$, let $K, L \subset \mathbb{R}^n$ be full-dimensional origin-symmetric convex bodies, and let $0 < p < 1$. For every $0 \leq \lambda \leq 1$,*

$$|\mathcal{W}[(1-\lambda)h_K^p + \lambda h_L^p]^{1/p}|^{p/n} \geq (1-\lambda)|K|^{p/n} + \lambda|L|^{p/n}. \quad (1.2)$$

The proof of Corollary 1.2 is given in Section 8.1.

The logarithmic volume theorem has a separate measure-theoretic consequence. An even log-concave density has the form e^{-V} , where V is an even convex function with values in $\mathbb{R} \cup \{+\infty\}$; no probability normalization is required. Saroglou proved that the Lebesgue inequality in dimension n implies the same Wulff inequality for every such density in that same dimension [29, Theorem 3.1]. Taking two scalar dilates of a single body then gives the (B) -conjecture: for the measure $d\mu = e^{-V(x)} dx$ and every origin-symmetric convex body K , the function $t \mapsto \mu(e^t K)$ is log-concave on $\mathbb{R}$ [29, Corollary 3.2]. Section 8.2 applies this transfer and also treats even log-concave Radon measures supported on proper subspaces.

Context and prior work

Theorem 1.1 resolves the origin-symmetric logarithmic Brunn–Minkowski conjecture of Böröczky, Lutwak, Yang and Zhang [6, Problem 1.1]. Stancu [30, Theorem 2, p. 3264] announced the all-dimensional logarithmic Minkowski inequality and explicitly concluded the equivalent logarithmic Brunn–Minkowski inequality; the report deferred details of the flow asymptotics. The conjecture belongs to the L_p Brunn–Minkowski theory initiated by Firey’s p -means of convex bodies [17] and developed by Lutwak through L_p mixed volumes and the L_p Minkowski problem [24]. For $p > 0$, the support bounds are interpolated by the weighted power mean appearing in Corollary 1.2; the geometric mean is its limit as $p \downarrow 0$. This endpoint has a particularly close connection with cone-volume measures and the logarithmic Minkowski problem. The cone-volume measure records the

volumes of cones from the origin to boundary patches, indexed by their outer normals. Böröczky, Lutwak, Yang and Zhang established the equivalence between the corresponding logarithmic volume and mixed-volume inequalities [6, Lemma 3.2]; their subsequent work characterized which even measures arise as cone-volume measures [7, Theorem 1.1]. The latter is an existence result, separate from the volume inequality proved here.

Several important classes were established in earlier work. Böröczky, Lutwak, Yang and Zhang proved the planar case [6, Theorem 1.3]. Saroglou established the inequality and studied its equality cases for bodies unconditional with respect to the same orthonormal basis, meaning that both bodies are invariant under every coordinate sign change in that basis [28, Theorem 1.2]. The underlying volume inequality for coordinatewise products has antecedents in Uhrin, Bollobás–Leader, and Cordero-Erausquin–Fradelizi–Maurey [31, 3, 11]; Saroglou connected this product to the logarithmic combination. This argument uses the multiplicative form of the Prékopa–Leindler inequality. Böröczky and Kalantzopoulos extended the symmetry method to bodies invariant under the same n linear reflections whose fixed hyperplanes intersect only at the origin [5, Theorem 2]. For unit balls of complex norms, Rotem derived the inequality from Cordero-Erausquin’s volume inequality for complex interpolation [27, 12].

A complementary approach studies the second variation of volume. Colesanti, Livshyts and Marsiglietti established a local result near Euclidean balls and developed infinitesimal formulations [10]. Kolesnikov and Milman expressed the local L_p inequality as a lower bound for the first nonconstant even eigenvalue of the Hilbert–Brunn–Minkowski operator, and proved it for $p \geq 1 - cn^{-3/2}$ with a universal $c > 0$ [22]. Their spectral formulation explains why symmetry matters: the translation eigenmodes are odd and hence disappear in the even subspace. Chen, Huang, Li and Liu obtained the corresponding global inequalities for p sufficiently close to 1 by a PDE argument [9]. Putterman proved the equivalence of local and global L_p inequalities for every $p \in [0, 1)$ by studying strongly isomorphic polytopes [26, Theorem 1.7]. Van Handel proved the local logarithmic inequality for origin-symmetric zonoids, that is, Hausdorff limits of sums of segments, against arbitrary origin-symmetric comparison bodies [32, Theorem 1.4 of the cited arXiv version]. Milman’s centro-affine interpretation of the operator led to further curvature-dependent results and an isomorphic logarithmic Minkowski theorem: every symmetric body has a suitable replacement within a universal geometric factor [25]. These developments identify the analytic obstruction without imposing a common normal fan on the global conjecture.

Iffland gave another spectral proof of the local logarithmic inequality for origin-symmetric bodies of revolution and extended it to nonsymmetric comparison bodies under a weighted centering condition [18, Theorem A]. Lu proved the global inequality for origin-symmetric pairs sharing $n - 2$ orthogonal reflection symmetries [23, Theorems 1.2 and 1.3]. Xi [34, Theorem 1.1] gives a new planar proof that also treats Dar’s conjecture. These results illustrate the different roles of local variation and additional symmetries in the problem.

Proof strategy

For finitely many spanning unit vectors p_i and positive numbers h_i , the constraints $|\langle p_i, x \rangle| \leq h_i$ define a symmetric polytope. Replace its indicator by the smooth weight

$$\exp \left(-\frac{\varepsilon}{2}|x|^2 - \sum_i \left(\frac{\langle p_i, x \rangle}{h_i} \right)^q \right), \quad \varepsilon > 0, \quad q \geq 2 \text{ even}.$$

As q tends to infinity, this weight becomes the Gaussian weight restricted to the polytope. Letting ε tend to zero recovers its volume. Thus it suffices to prove that the integral of the smooth weight

is log-concave when each h_i varies geometrically. Section 2 makes this reduction precise, including the passage from finitely many directions to all directions.

Geometric changes of slab widths are also related to the (B) -property, which asks for log-concavity of measure under exponential dilations. Saroglou showed that the diagonal (B) -property for uniform measures on cubes in every dimension is equivalent to the logarithmic Brunn–Minkowski conjecture in every dimension [28, Theorem 1.5]. This relates the geometric reduction to an earlier measure-theoretic formulation. The diagonal statement here is restricted to uniform measures on cubes; the consequence for general even log-concave measures concerns scalar dilations. The summandwise variance estimate below supplies the required concavity directly.

The second derivative of the logarithm of that integral is a variance minus an expected second derivative. The analytic task is therefore a variance bound for sums of even powers of linear forms (Proposition 2.1). We prove it in moment coordinates: the probability with density proportional to the displayed weight is the image of $e^{-\varphi(z)} dz$ under $x = \nabla\varphi(z)$. Section 3 constructs these coordinates with a bounded, everywhere positive Hessian, starting from the compact moment theorem of Berman and Berndtsson [2] and Klartag’s Hessian estimate [19]. The general moment-measure theory is due to Cordero-Erausquin and Klartag [13].

Two steps complete the analytic argument. First, a differential operator associated with the moment Hessian produces a dense class of centered even test functions (Sections 4 and 5). The only obstructions to density are affine functions: centering removes constants, and evenness removes linear functions. Second, a tensor calculation for these tests has a remainder that is the sum of explicit squares (Section 6). This supplies the variance bound by two Cauchy–Schwarz inequalities (Section 7).

The analytic setting belongs to the tradition of Brascamp–Lieb variance inequalities for log-concave measures [8]. Kolesnikov, Livshyts and Rotem [20, Theorem 1.4 and Corollaries 1.5–1.6] relate strengthened Brascamp–Lieb inequalities for homogeneous density potentials to local p -Brunn–Minkowski inequalities and obtain the local logarithmic inequality for ℓ_q balls, $q \geq 1$. Their homogeneous variance formulation uses the Hessian of the density potential; our summandwise estimate uses the separately constructed moment potential. The transport-Hessian diffusion and curvature formulas were studied earlier by Kolesnikov [21]; the Bochner identities in [20, Section 4.3] build on this framework.

The ingredient here is a tensor estimate in moment coordinates that controls each homogeneous summand separately. The accompanying density argument uses a global upper Hessian bound and local strict positivity; it does not require a global lower Hessian bound. We state these two steps separately so that their hypotheses and their possible use for other variance estimates are explicit.

2 From a variance bound to volume

We first deduce Theorem 1.1 from the variance bound below. For a probability measure μ , write $\mathbb{E}_\mu f = \int f d\mu$ and $\text{Var}_\mu(f) = \mathbb{E}_\mu(f - \mathbb{E}_\mu f)^2$.

Proposition 2.1 (Variance bound). *Let $n \geq 2$, $N \geq 1$, $q \geq 2$ be an even integer, and $\varepsilon > 0$. Let $p_1, \dots, p_N \in \mathbb{R}^n \setminus \{0\}$ and $h_1, \dots, h_N > 0$. Set*

$$\psi_i(x) = \left(\frac{\langle p_i, x \rangle}{h_i} \right)^q, \quad V(x) = \frac{\varepsilon}{2} |x|^2 + \sum_{i=1}^N \psi_i(x), \quad d\mu = Z^{-1} e^{-V(x)} dx, \quad (2.1)$$

where $Z = \int_{\mathbb{R}^n} e^{-V(x)} dx$. For all real $b_1, \dots, b_N$,

$$\text{Var}_\mu \left(\sum_{i=1}^N b_i \psi_i \right) \leq \sum_{i=1}^N b_i^2 \mathbb{E}_\mu \psi_i. \quad (2.2)$$

Here the Gaussian term makes Z finite, even if the vectors p_i do not span $\mathbb{R}^n$. All polynomial moments are finite. The proof of Proposition 2.1 occupies Sections 3–7.

The proof will establish the dual estimate

$$\sum_{i=1}^N \frac{(\mathbb{E}_\mu u \psi_i)^2}{\mathbb{E}_\mu \psi_i} \leq \mathbb{E}_\mu u^2 \quad \text{for every centered even } u \in L^2(\mu).$$

Here centered means $\mathbb{E}_\mu u = 0$, and the denominators are positive because each $p_i \neq 0$ and μ has a positive density. Applied to $u = \sum_i b_i \psi_i - \mathbb{E}_\mu \sum_i b_i \psi_i$, this gives (2.2) by Cauchy–Schwarz. We first prove the dual estimate on a class of smooth compactly supported tests, then use density to reach every such u . This explains the roles of the density and tensor lemmas in the analytic proof.

Proof of Theorem 1.1 from Proposition 2.1. Suppose first that $n \geq 2$. Fix nonzero spanning vectors $p_1, \dots, p_N$ and positive endpoint widths h_i^0, h_i^1 . For $t \in [0, 1]$, define

$$h_i(t) = (h_i^0)^{1-t} (h_i^1)^t, \quad P(t) = \{x : |\langle p_i, x \rangle| \leq h_i(t) \text{ for all } i\}.$$

The spanning condition makes each $P(t)$ bounded; positivity of the widths gives nonempty interior. Use these widths in (2.1) and denote the resulting integral by $Z_{q,\varepsilon}(t)$. For fixed q, ε , its first two derivatives may be taken under the integral: on a compact interval of t , the differentiated integrands are bounded by a polynomial times $e^{-\varepsilon|x|^2/2}$. With $b_i = -q \log(h_i^1/h_i^0)$ one has

$$\dot{V} = \sum_i b_i \psi_i, \quad \ddot{V} = \sum_i b_i^2 \psi_i,$$

and hence

$$\frac{d^2}{dt^2} \log Z_{q,\varepsilon}(t) = \text{Var}_{\mu_t}(\dot{V}) - \mathbb{E}_{\mu_t} \ddot{V} \leq 0$$

by Proposition 2.1. Therefore

$$Z_{q,\varepsilon}(\lambda) \geq Z_{q,\varepsilon}(0)^{1-\lambda} Z_{q,\varepsilon}(1)^\lambda. \quad (2.3)$$

Keep $\varepsilon > 0$ fixed and let q tend to infinity through even integers. For each fixed t , the integrands converge almost everywhere to $e^{-\varepsilon|x|^2/2} \mathbf{1}_{P(t)}$. The exceptional set lies in the finitely many hyperplanes $\langle p_i, x \rangle = \pm h_i(t)$ and has measure zero. The common integrable bound $e^{-\varepsilon|x|^2/2}$ gives

$$Z_{q,\varepsilon}(t) \longrightarrow \int_{P(t)} e^{-\varepsilon|x|^2/2} dx.$$

Apply this at $t = 0, \lambda, 1$ in (2.3), then let $\varepsilon \downarrow 0$. Monotone convergence gives

$$|P(\lambda)| \geq |P(0)|^{1-\lambda} |P(1)|^\lambda. \quad (2.4)$$

Figure 1 illustrates how additional slab directions refine an outer approximation. To pass to all directions, let K, L be as in Theorem 1.1. Choose nested finite sets $D_m \subset S^{n-1}$ that contain the coordinate vectors and whose union is dense in the sphere. Define

$$P_m(t) = \bigcap_{p \in D_m} \{x : |\langle p, x \rangle| \leq h_K(p)^{1-t} h_L(p)^t\}.$$

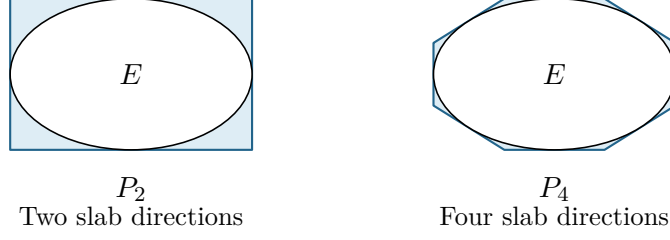


Figure 1: Finite outer approximations of an ellipse E . Adding slab directions gives $P_2 \supset P_4 \supset E$; the subscripts here count slab directions. Using nested finite spanning direction sets with dense union, the same construction recovers $\mathcal{W}[f]$ from any positive continuous even function f on the sphere; f need not be a support function.

The bodies $P_m(t)$ decrease with m and are all contained in the bounded body $P_1(t)$. By continuity of the support functions and their evenness,

$$\bigcap_m P_m(t) = \mathcal{W}[h_K^{1-t} h_L^t].$$

Indeed any point satisfying the dense set of inequalities satisfies all of them by continuity in the direction. Conversely, the Wulff inequalities imply the absolute-value constraints by evenness. At $t = 0, 1$ these intersections are K, L , respectively, by the supporting-half-space characterization of a convex body. Continuity of measure from above, applied at $0, \lambda, 1$ in (2.4), proves (1.1).

Finally, in dimension one write $K = [-a, a]$ and $L = [-b, b]$ with $a, b > 0$. Their Wulff interpolation is the interval with radius $a^{1-\lambda}b^\lambda$, whose length is $(2a)^{1-\lambda}(2b)^\lambda$. Thus (1.1) is equality. The endpoints $\lambda = 0, 1$ are equality in every dimension. $\square$

3 Moment coordinates on the whole space

The variance argument uses a change of variables whose Jacobian is a positive Hessian. The following lemma supplies this change of variables and the uniform upper bound needed for the later cutoff arguments.

Lemma 3.1 (Moment coordinates). *Let $n \geq 2$, $\varepsilon > 0$, and let $V \in C^\infty(\mathbb{R}^n)$ be even with $D^2V \geq \varepsilon I$. Set*

$$Z = \int_{\mathbb{R}^n} e^{-V(x)} dx, \quad d\mu(x) = Z^{-1} e^{-V(x)} dx.$$

There is an even $\varphi \in C^\infty(\mathbb{R}^n)$ such that $d\nu(z) = e^{-\varphi(z)} dz$ is a probability measure, $\nabla\varphi$ is a diffeomorphism of $\mathbb{R}^n$ onto $\mathbb{R}^n$, and

$$(\nabla\varphi)_\# \nu = \mu, \quad 0 < D^2\varphi \leq \frac{4}{\varepsilon} I, \quad \log \det D^2\varphi = V(\nabla\varphi) + \log Z - \varphi. \quad (3.1)$$

Moreover, $\varphi(z) \geq c|z| - b$ for some $c > 0$ and $b < \infty$. The lower Hessian bound in (3.1) means positive definiteness at every point; it need not be uniform on $\mathbb{R}^n$.

Proof. Evenness and uniform convexity give $V(x) \geq V(0) + \varepsilon|x|^2/2$, so $0 < Z < \infty$. For $R \geq 2$, write $B_R = \{x : |x| < R\}$ and set

$$Z_R = \int_{B_R} e^{-V(x)} dx, \quad d\mu_R = Z_R^{-1} \mathbf{1}_{B_R} e^{-V(x)} dx.$$

We first use two compact-support results. The moment theorem of Berman–Berndtsson [2, Theorem 1.1], applied to the body $\bar{B}_R$ and the positive smooth density e^{-V}/Z_R , gives a smooth convex potential φ_R , unique up to translation, for which $\nabla\varphi_R : \mathbb{R}^n \rightarrow B_R$ is a diffeomorphism and

$$\det D^2\varphi_R = Z_R \exp\{V(\nabla\varphi_R) - \varphi_R\}. \quad (3.2)$$

The centering hypothesis holds by evenness. Integrating the equation through this diffeomorphism shows that $d\nu_R = e^{-\varphi_R} dz$ is a probability and $(\nabla\varphi_R)_\# \nu_R = \mu_R$. Klartag’s estimate [19, Proposition 3.1 and Remark 3.5] gives

$$0 < D^2\varphi_R \leq CI, \quad C = 4/\varepsilon.$$

Indeed, the ball has smooth positively curved boundary, and $\rho_R = V + \log Z_R$ is convex and smooth, with itself and all its derivatives bounded on B_R , and $D^2\rho_R \geq \varepsilon I$. The constant C is independent of R .

Translate the unique preimage of 0 to 0. It is the unique minimum of φ_R . Reflection gives a second potential for the same target with its minimum at 0; uniqueness up to translation therefore makes φ_R even. For each fixed R , take the preimages of the slopes $\pm(R/2)e_i$. Their supporting planes give

$$\varphi_R(z) \geq \frac{R}{2\sqrt{n}}|z| - b_R$$

with a finite, possibly R -dependent constant b_R . Thus ν_R has an exponential tail. In particular, since $|\nabla\varphi_R| \leq R$, integration of $\operatorname{div}(ze^{-\varphi_R})$ with expanding compact cutoffs is justified and yields

$$\mathbb{E}_{\nu_R} z \cdot \nabla\varphi_R(z) = n. \quad (3.3)$$

We now obtain bounds independent of R . Put $m_R = \varphi_R(0)$ and $g_R = \varphi_R - m_R \geq 0$. The Hessian bound implies $g_R(y) \leq C|y|^2/2$ and $|\nabla\varphi_R(y)| \leq C|y|$. For arbitrary y, z , set $x = \nabla\varphi_R(z)$. Supporting planes at z and $-z$ give the single pointwise inequality

$$g_R(y) \geq g_R(z) + |x \cdot y| - x \cdot z.$$

Integrate with respect to ν_R , discard the nonnegative $\mathbb{E}_{\nu_R} g_R$, and use (3.3):

$$g_R(y) \geq \int |x \cdot y| d\mu_R(x) - n.$$

On B_1 the density of μ_R is at least $d_0 = Z^{-1} \exp(-\max_{\bar{B}_1} V) > 0$. Rotational symmetry of the ball therefore bounds the integral below by $c|y|$, where $c = d_0 \int_{B_1} |x_1| dx > 0$. Hence

$$c|y| - n \leq g_R(y) \leq \frac{C}{2}|y|^2, \quad \left(\frac{2\pi}{C}\right)^{n/2} \leq e^{m_R} = \int e^{-g_R(y)} dy \leq e^n \int e^{-c|y|} dy. \quad (3.4)$$

This bounds m_R on both sides and gives one integrable exponential majorant for all the densities $e^{-\varphi_R}$.

It remains to obtain a smooth limit and to verify that its gradient reaches every point. On each fixed ball, the preceding estimates bound φ_R and $\nabla\varphi_R$ uniformly. Since $Z_2 \leq Z_R \leq Z$, equation (3.2) bounds $\det D^2\varphi_R$ below by a positive constant there. Together with $D^2\varphi_R \leq CI$, this gives a uniform local bound $\delta I \leq D^2\varphi_R \leq CI$: if the determinant is at least $d > 0$, every eigenvalue is at least d/C^{n-1} . The right-hand sides

$$f_R = V(\nabla\varphi_R) + \log Z_R - \varphi_R$$

of the logarithmic equations are uniformly Lipschitz on that ball, because $\nabla f_R = D^2\varphi_R \nabla V(\nabla\varphi_R) - \nabla\varphi_R$ is uniformly bounded.

To apply an interior uniformly elliptic estimate, choose a smooth concave scalar function s equal to log on a slightly larger interval than $[\delta, C]$, with s' bounded above and bounded below by positive constants on $\mathbb{R}$. Such an extension is obtained by smoothly continuing the decreasing derivative $1/t$ to positive constant tails. For symmetric matrices Q , the spectral function $\mathcal{F}(Q) = \text{tr } s(Q)$ is smooth, concave and uniformly elliptic: its derivative has eigenvalues $s'(\lambda_i(Q))$, and its second variation has coefficients $s''(\lambda_i)$ and the nonpositive divided differences of s' . It agrees with log det on the Hessians under consideration. The nonhomogeneous Evans–Krylov estimate [33, Corollary 2.3] applied to $\mathcal{F}(D^2\varphi_R) = f_R$ gives uniform interior $C^{2,\alpha}$ bounds for some $\alpha > 0$.

Each φ_R is already smooth. Its derivative $w_{R,k} = \partial_k \varphi_R$ satisfies

$$(D^2\varphi_R)^{-1} : D^2w_{R,k} = \nabla V(\nabla\varphi_R) \cdot \nabla w_{R,k} - w_{R,k},$$

where $Q : N = \text{tr}(Q^T N)$. The principal coefficients and the right-hand side are uniformly C^α on smaller balls. Interior Schauder estimates [16, Theorem 2.20 and Corollary 2.21] give uniform $C^{3,\alpha}$ bounds for φ_R , and iteration gives bounds for every derivative on compact sets. A diagonal subsequence, still indexed by R , converges locally smoothly to an even φ with positive definite Hessian, the same upper cap, and the logarithmic equation in (3.1). The bound $\varphi_R(z) \geq c|z| - b$, with b independent of R , also passes to φ .

The common exponential majorant from (3.4) proves $\int e^{-\varphi} = 1$ by dominated convergence. For each bounded continuous h , it also gives

$$\int h(\nabla\varphi) e^{-\varphi} = \lim_{R \rightarrow \infty} \int h(\nabla\varphi_R) e^{-\varphi_R} = \lim_{R \rightarrow \infty} \int h d\mu_R = \int h d\mu.$$

Thus $(\nabla\varphi)_\# \nu = \mu$. Positive definite Hessian makes the gradient locally invertible and injective, since for $z \neq w$,

$$(\nabla\varphi(z) - \nabla\varphi(w)) \cdot (z - w) = \int_0^1 D^2\varphi(w + t(z - w)) [z - w, z - w] dt > 0.$$

Its image is dense: an open ball missed by the image would have zero pushforward mass, contrary to the everywhere positive density of μ . Fix $x_0 \in \mathbb{R}^n$. Density permits finitely many image points $y_j = \nabla\varphi(z_j)$ whose convex hull contains a ball $B_r(x_0)$ with $r > 0$; one may take small perturbations of the points $x_0 \pm e_i$. Their supporting planes imply

$$\varphi(z) - x_0 \cdot z \geq \max_j \{(y_j - x_0) \cdot z + \varphi(z_j) - y_j \cdot z_j\} \geq r|z| - b_{x_0}.$$

This function attains a minimum, at which $\nabla\varphi = x_0$. The gradient is therefore onto, and its local smooth inverses combine to a global smooth inverse. $\square$

We will write $x = \nabla\varphi(z)$ and $\tau = D_z^2\varphi$, evaluating τ at the corresponding point when using x coordinates. Thus $\tau(x) = D_z^2\varphi((\nabla\varphi)^{-1}(x))$ is smooth, even, positive definite and bounded above by $(4/\varepsilon)I$. The gradient is odd, as is its inverse. Finally, a compactly supported smooth function of x is compactly supported after composition with $\nabla\varphi$: its support lies in the image of a compact set under the continuous inverse map. Consequently the compact integrations by parts below are valid in either coordinate system.

4 Adjoints in moment coordinates

We retain the probability measures

$$d\mu(x) = Z^{-1}e^{-V(x)} dx, \quad d\nu(z) = e^{-\varphi(z)} dz, \quad x = \nabla\varphi(z),$$

and the smooth diffeomorphism constructed above. Put

$$\tau = D_z^2\varphi, \quad M = \tau^{-1}.$$

In target coordinates, these symbols denote the same matrix fields evaluated at $z = (\nabla\varphi)^{-1}(x)$. Expectation under either measure, with functions identified by this map, is denoted by $\mathbb{E}$. Throughout the coordinate calculations, repeated indices are summed from 1 to n , and $Q : L = \text{tr}(Q^T L)$ denotes matrix contraction.

We first construct the compact test functions used in the variance argument and establish an identity that will also identify the obstruction to their density. Write

$$\partial_k = \partial_{z_k}, \quad \partial_k^* = x_k - \partial_k.$$

The latter is the adjoint of ∂_k in $L^2(\nu)$ because $\partial_k\varphi = x_k$. In target coordinates, write δ_μ for negative weighted divergence:

$$\delta_\mu W = \sum_j (V_{x_j} W_j - \partial_{x_j} W_j).$$

On a matrix, δ_μ acts on each row, producing a vector. For a smooth row field P_{ak} and a compact test g , the chain rule and adjunction give

$$\mathbb{E}g \partial_k^* P_{ak} = \mathbb{E}(\partial_k g) P_{ak} = \mathbb{E}g_{x_i} \tau_{ik} P_{ak}.$$

Consequently

$$\partial_k^* P_{ak} = (\delta_\mu(P\tau))_a. \tag{4.1}$$

Taking P to be the identity matrix also gives $\delta_\mu\tau = x$. This is Fathi's moment-map construction of a Stein kernel [15, Theorem 2.3 of the cited arXiv version]: the matrix field τ expresses integration by parts against x through $\mathbb{E}x_j g = \mathbb{E}\tau_{jk} g_{x_k}$.

Thus the differential operator

$$Af = \delta_\mu(\tau \nabla_x f) = \partial_k^* f_{x_k} = x \cdot \nabla_x f - \tau : D_x^2 f \tag{4.2}$$

is formally symmetric, with

$$\mathbb{E}g Af = \mathbb{E}\nabla_x g \cdot \tau \nabla_x f \tag{4.3}$$

for compact smooth tests. These formulas involve no operator-domain assertion. This is the negative of the moment-Hessian diffusion generator studied by Klartag [19, Section 6]; for our sign convention, $Ax_j = x_j$.

For $f \in C_c^\infty(\mathbb{R}_x^n)$, define

$$h = D_x^2 f, \quad P = \tau h, \quad B = \tau h \tau, \quad W_a = \partial_k^* P_{ak}, \quad u = \delta_\mu W. \tag{4.4}$$

All these fields have compact support in both coordinates: the inverse image of a compact target set is its compact image under the continuous inverse diffeomorphism. By (4.1), $W = \delta_\mu B$. For every smooth function F in target coordinates, two adjunctions give

$$\mathbb{E}uF = \mathbb{E}W \cdot \nabla_x F = \mathbb{E}B : D_x^2 F. \tag{4.5}$$

The fields are compactly supported, so F need not be. Thus the same pairing can be estimated through either the first or the second derivatives of F . To identify the range of these tests, we also use the gradient identity

$$W = \nabla_z [((A-1)f)(x(z))] = \tau \nabla_x (A-1)f, \quad u = A(A-1)f. \quad (4.6)$$

To check it directly, put $C_{ijk} = \partial_{z_i} \partial_{z_j} \partial_{z_k} \varphi$. This tensor is fully symmetric, and expansion gives

$$W_a = x_k \tau_{ai} f_{x_i x_k} - C_{aik} f_{x_i x_k} - \tau_{ai} \tau_{jk} f_{x_i x_k x_j}.$$

Differentiating $(A-1)f$ in z_a gives exactly these terms: the first-gradient terms from $x \cdot \nabla_x f$ and $-f$ cancel. The formula for u follows from (4.2). In particular, $D_z W$ is symmetric.

The following identity is the moment-coordinate specialization of the transport-Hessian Bochner formula of Kolesnikov, Livshyts and Rotem [20, Proposition 4.8]. We give its direct derivation in the present notation. Two compact integrations by parts give

$$\begin{aligned} \mathbb{E}(Af)(A-1)f &= \mathbb{E} \nabla_x f \cdot \tau \nabla_x (A-1)f = \mathbb{E} \nabla_x f \cdot W \\ &= \mathbb{E} h : B = \mathbb{E} \|\tau^{1/2} D_x^2 f \tau^{1/2}\|_{\text{HS}}^2. \end{aligned} \quad (4.7)$$

Here $\|Q\|_{\text{HS}}^2 = \text{tr}(Q^T Q)$. The differential expression A satisfies $A1 = 0$ and $Ax_j = x_j$, so $A(A-1)$ annihilates every affine function. Section 5 uses (4.7) to prove that these are the only L^2 obstructions to density. Centering removes constants, and evenness removes linear functions.

5 Density of the even tests

The Hessian identity identifies affine functions as the possible obstructions to density. We now show that these are the only obstructions, using cutoffs in the x coordinates.

Lemma 5.1 (Density of even tests). *Let $d\mu = \rho(x) dx$ be a probability measure on $\mathbb{R}^n$ with a smooth, strictly positive, even density. Let τ be a smooth even symmetric matrix field with $0 < \tau(x) \leq \Lambda I$ for some finite Λ , and set*

$$Af = x \cdot \nabla f - \tau : D^2 f.$$

Suppose that A has the integration formula

$$\mathbb{E}[hAf] = \mathbb{E}[\nabla h \cdot \tau \nabla f] \quad (f, h \in C_c^\infty(\mathbb{R}^n))$$

and satisfies the compact Hessian identity (4.7). Then

$$\overline{\{A(A-1)f : f \in C_c^\infty(\mathbb{R}^n), f \text{ even}\}}^{L^2(\mu)} = \{g \in L^2(\mu) : g \text{ even}, \mathbb{E}g = 0\}.$$

Proof. Write $\Gamma(f, h) = \nabla f \cdot \tau \nabla h$ and $\Gamma(f) = \Gamma(f, f)$. The integration formula also holds when one factor is smooth and the other has compact support, by inserting a cutoff around that support. In particular $\mathbb{E}Ak = 0$ for compactly supported smooth k . Since A preserves parity, the displayed range consists of centered even functions.

Let g be a centered even $L^2(\mu)$ function orthogonal to this range. It also annihilates $A(A-1)f$ for odd f , because that image is odd. Formal symmetry therefore gives

$$\mathcal{P}g = 0 \quad \text{in distributions,} \quad \mathcal{P} = A(A-1).$$

Indeed, if $\mathcal{P}^t$ denotes the transpose with respect to Lebesgue measure and $d\mu = \rho dx$, weighted formal symmetry gives $\mathcal{P}^t(\rho g) = \rho \mathcal{P}g$ in distributions. Orthogonality makes the left side zero, and smooth positivity of ρ gives the displayed equation. The principal symbol of $\mathcal{P}$ is $(\xi \cdot \tau(x)\xi)^2$, which is positive for every $\xi \neq 0$. Interior elliptic regularity consequently gives $g \in C^\infty$ [14, Theorem 14.2]. Only positivity on compact sets is needed here.

Splitting the equation in L^2 . Choose $0 \leq \eta_r \leq 1$ smooth, equal to one on B_r , zero outside B_{2r} , with $|\nabla \eta_r| \leq C/r$ and $|D^2 \eta_r| \leq C/r^2$, where $r \geq 1$. The upper bound on τ gives constants B, K , independent of r , such that

$$\Gamma(\eta_r)^{1/2} \leq B/r, \quad |A\eta_r| \leq K. \quad (5.1)$$

For the second estimate, use $|x| \leq 2r$ on the support of the cutoff derivatives. Those derivatives, including $A\eta_r$, vanish outside $B_{2r} \setminus B_r$.

Put $v = Ag$, initially just a smooth function, and write

$$G = \|g\|_2, \quad X_r = \|\eta_r^2 v\|_2, \quad Y_r^2 = \mathbb{E}[\eta_r^2 \Gamma(g)].$$

Here G is finite by hypothesis, while X_r, Y_r are finite by compact support. Testing $\mathcal{P}g = 0$ against $\eta_r^4 g$ gives $\mathbb{E}[v(A-1)(\eta_r^4 g)] = 0$. The product rule $A(ab) = aAb + bAa - 2\Gamma(a, b)$ yields the exact expansion

$$(A-1)(\eta_r^4 g) = \eta_r^4 v + g(4\eta_r^3 A\eta_r - 12\eta_r^2 \Gamma(\eta_r) - \eta_r^4) - 8\eta_r^3 \Gamma(\eta_r, g).$$

Thus Cauchy–Schwarz and (5.1) imply

$$X_r \leq aG + \frac{8B}{r}Y_r, \quad Y_r^2 \leq GX_r + \frac{2B}{r}GY_r, \quad a = 1 + 4K + 12B^2. \quad (5.2)$$

The first inequality is immediate if $X_r = 0$; otherwise divide the resulting estimate for X_r^2 by X_r . For the second, integrate by parts in $\mathbb{E}[\eta_r^2 g Ag]$:

$$Y_r^2 = \mathbb{E}[\eta_r^2 gv] - 2\mathbb{E}[\eta_r g \Gamma(\eta_r, g)].$$

Combining (5.2) gives $Y_r^2 \leq aG^2 + 10BGY_r$, so both Y_r and X_r are bounded uniformly in r . Exhaustion by balls proves $v = Ag \in L^2(\mu)$. We may now set

$$w = g - v \in L^2(\mu), \quad Aw = 0, \quad Av = v.$$

The harmonic part is constant. Testing $Aw = 0$ with $\eta_r^2 w$ and using (5.1) gives

$$(\mathbb{E}[\eta_r^2 \Gamma(w)])^{1/2} \leq \frac{2B}{r} \|w\|_2.$$

On any fixed ball, let $r \rightarrow \infty$. Then $\Gamma(w) = 0$ there; strict positivity of τ shows that w is constant.

The eigenvalue-one part is linear. The analogous test of $Av = v$ gives, with $Z_r^2 = \mathbb{E}[\eta_r^2 \Gamma(v)]$,

$$Z_r^2 \leq \|v\|_2^2 + \frac{2B}{r} \|v\|_2 Z_r.$$

Hence $\mathbb{E}\Gamma(v) < \infty$. For $f_r = \eta_r v \in C_c^\infty$, the product rule now proves directly that

$$f_r \longrightarrow v, \quad Af_r \longrightarrow v \quad \text{in } L^2(\mu). \quad (5.3)$$

Indeed,

$$Af_r = \eta_r v + v A \eta_r - 2\Gamma(\eta_r, v).$$

The middle term has norm at most $K\|v\mathbf{1}_{B_{2r}\setminus B_r}\|_2 \rightarrow 0$. For the last term, use

$$\|\Gamma(\eta_r, v)\|_2 \leq \frac{B}{r}(\mathbb{E}\Gamma(v))^{1/2} \rightarrow 0.$$

Apply (4.7) to these compactly supported functions:

$$\mathbb{E}[\|\tau^{1/2}D^2f_r\tau^{1/2}\|_{\text{HS}}^2] = \|Af_r\|_2^2 - \mathbb{E}[f_r Af_r] \rightarrow 0.$$

On each fixed ball, $f_r = v$ for all sufficiently large r . Positivity of the density and of τ therefore forces $D^2v = 0$ on that ball. Thus v is affine, and $Av = v$ removes its constant term.

We have proved that every L^2 distributional solution of $\mathcal{P}g = 0$ is affine. Our $g = w + v$ is even, so it is constant, and its mean is zero. Hence $g = 0$. Taking the orthogonal complement in the centered even subspace proves the lemma. $\square$

6 A tensor estimate for the divergence tests

The gradient and Hessian pairings in (4.5) will be estimated using two quadratic energies, one for W and one for B . The following lemma bounds their sum by $\mathbb{E}u^2$. Its proof uses weighted integration by parts and the moment equation; homogeneity enters only when the lemma is applied to the summands ψ_i in Section 7.

Lemma 6.1 (Moment-coordinate tensor estimate). *Let $d\mu = Z^{-1}e^{-V}dx$ and $d\nu = e^{-\varphi}dz$ be smooth probability measures on $\mathbb{R}^n$. Suppose $D_z^2\varphi$ is positive definite and $x = \nabla\varphi(z)$ is a smooth diffeomorphism pushing ν to μ . For a real $f \in C_c^\infty(\mathbb{R}_x^n)$, put*

$$\tau = D_z^2\varphi, \quad M = \tau^{-1}, \quad H = D_x^2V, \quad B = \tau(D_x^2f)\tau, \quad W = \delta_\mu B, \quad u = \delta_\mu W.$$

View source fields in target coordinates by the inverse moment map. Then

$$\mathbb{E} \operatorname{tr}((D_x W)^2) \geq \mathbb{E} \operatorname{tr}(MBHB), \tag{6.1}$$

and

$$\mathbb{E}u^2 \geq \mathbb{E} \operatorname{tr}(MBHB) + \mathbb{E}W^T HW. \tag{6.2}$$

Here $(D_x W)_{ij} = \partial_{x_j} W_i$, and the square in (6.1) is a matrix square.

Proof. All test fields used in the integrations by parts below have compact support. Weighted Bochner calculations underlie many variance estimates; see Bakry–Émery [1, Proposition 3] and the transport-Hessian formulas in [20, Section 4.3]. Here the required identity is for a vector field, so we derive it explicitly and retain the matrix trace $\operatorname{tr}((D_x W)^2)$. First, for any such vector field W and $u = \delta_\mu W$,

$$\partial_{x_i} u = H_{ij} W_j + (V_{x_j} - \partial_{x_j}) \partial_{x_i} W_j.$$

Pairing with W_i and using adjunction gives the weighted identity

$$\mathbb{E}u^2 = \mathbb{E}W_i \partial_{x_i} u = \mathbb{E}W^T HW + \mathbb{E}(\partial_{x_j} W_i)(\partial_{x_i} W_j) = \mathbb{E}W^T HW + \mathbb{E} \operatorname{tr}((D_x W)^2). \tag{6.3}$$

Thus it remains to prove (6.1). Its proof compares two expansions, integrates their derivative terms by parts, and writes the resulting difference as a sum of squares.

Recall $h = D_x^2 f$ and $P = \tau h$ from (4.4), and let $(C_i)_{kl} = C_{ikl}$, where $C = D_z^3 \varphi$. The moment equation (3.1) reads

$$V(x(z)) = \varphi(z) + \log \det \tau(z) - \log Z.$$

Its first derivative gives

$$\tau_{ik} V_{x_k} = x_i + c_i, \quad c_i = \text{tr}(MC_i).$$

Since $\partial_j M = -MC_j M$, differentiating once more gives

$$(\tau H \tau)_{ij} = \tau_{ij} + M_{kl} \varphi_{kl ij} - \text{tr}(MC_i MC_j) - C_{ijk} V_{x_k}.$$

To combine the fourth-derivative term and the term involving $\nabla_x V$ into a weighted divergence, note that

$$\partial_d M_{dk} = -M_{da} C_{abd} M_{bk} = -c_b M_{bk}, \quad \partial_d M_{dk} - x_d M_{dk} = -V_{x_k}.$$

Full symmetry of $D_z^4 \varphi$ therefore yields

$$(\tau H \tau - \tau)_{ij} = (\partial_d - x_d)(M_{dk} C_{ijk}) - \text{tr}(MC_i MC_j). \quad (6.4)$$

For the other expansion, the commutator $[\partial_l, \partial_k^*] = \tau_{kl}$ gives

$$D_z W = B + R, \quad R_{al} = \partial_k^* \partial_l P_{ak}.$$

Both $D_z W$ and B are symmetric, so R is symmetric. Since $D_x W = (B + R)M$ and $MBM = h$, expansion followed by one integration by parts gives

$$\mathbb{E} \text{tr}((D_x W)^2) = \mathbb{E} \text{tr}(MBMB) + 2\mathbb{E}(\partial_k h_{al})(\partial_l P_{ak}) + \mathbb{E} \text{tr}(MRMR). \quad (6.5)$$

The last term is nonnegative: $\text{tr}(MRMR) = \|M^{1/2} R M^{1/2}\|_{\text{HS}}^2$. On the right side of (6.1), the coefficient of $\tau H \tau$ in the trace contraction is

$$Q = MBMBM = h \tau h.$$

Substituting (6.4) and integrating its divergence term by parts gives

$$\mathbb{E} \text{tr}(MBHB) = \mathbb{E} \text{tr}(MBMB) - \mathbb{E}(\partial_d Q_{ij}) M_{dk} C_{ijk} - \mathbb{E} Q_{ij} \text{tr}(MC_i MC_j). \quad (6.6)$$

The common first term cancels. We now evaluate the three remaining contracted terms in coordinates where $\tau = I$ at the point under consideration.

This normalization uses a constant dual change of coordinates. For an invertible constant matrix T , set

$$z = T z', \quad x' = T^T x, \quad f'(x') = f(T^{-T} x').$$

The corresponding normalized potentials and partition function are

$$\varphi'(z') = \varphi(T z') - \log |\det T|, \quad V'(x') = V(T^{-T} x'), \quad Z' = |\det T| Z.$$

They preserve both probability measures by change of variables, $x' = \nabla_{z'} \varphi'$, and the moment equation. The chain rule gives

$$\begin{aligned} \tau' &= T^T \tau T, & B' &= T^T B T, & R' &= T^T R T, \\ M' &= T^{-1} M T^{-T}, & h' &= T^{-1} h T^{-T}, & H' &= T^{-1} H T^{-T}, \\ Q' &= T^{-1} Q T^{-T}, & P' &= T^T P T^{-T}, & W' &= T^T W. \end{aligned}$$

Each z -derivative index, including all three indices of C , transforms covariantly with T ; the same law holds for ∂_k^* . Also $D_{x'}W' = T^T(D_xW)T^{-T}$. It follows that every contracted scalar in (6.5) and (6.6) is invariant. For example, in $(\partial_k h_{al})(\partial_l P_{ak})$ each index pairs one covariant and one contravariant factor. At a fixed point we may therefore take $T = \tau^{-1/2}$, using the value of τ at that point. All derivatives are computed under this constant transformation. No moving frame is differentiated; the integrations by parts have already been completed.

In these normalized coordinates, put

$$S_{ijk} = f_{x_i x_j x_k}, \quad J_{ijk} = h_{ia} C_{ajk}.$$

The tensor S is fully symmetric, and J is symmetric in its last two indices. Write $J_{ijk}^{(12)} = J_{jik}$, and use the Euclidean tensor norm and inner product in this frame. The three contractions are

$$2(\partial_k h_{al})(\partial_l P_{ak}) = 2\|S\|^2 + 2\langle S, J \rangle, \quad (6.7)$$

$$(\partial_d Q_{ij})C_{ijd} = 2\langle S, J \rangle + \langle J, J^{(12)} \rangle, \quad (6.8)$$

$$Q_{ij} \operatorname{tr}(C_i C_j) = \|J\|^2. \quad (6.9)$$

For (6.7), use $\partial_k h_{al} = S_{alk}$ and $\partial_l P_{ak} = C_{ail} h_{ik} + S_{akl}$. For (6.8), differentiating the two factors h in $Q = h\tau h$ gives the two copies of $\langle S, J \rangle$. Differentiating its middle factor gives

$$\sum_{iabjd} h_{ia} C_{abd} h_{bj} C_{ijd} = \sum_d \operatorname{tr}(h C_d h C_d) = \langle J, J^{(12)} \rangle.$$

Finally, (6.9) follows by expanding $Q = h^2$ and the trace, using the full symmetry of C .

Thus, apart from the nonnegative R term, the integrand of (6.5) minus (6.6) is

$$2\|S\|^2 + 4\langle S, J \rangle + \|J\|^2 + \langle J, J^{(12)} \rangle.$$

Full symmetrization of J is

$$(\operatorname{Sym} J)_{ijk} = \frac{J_{ijk} + J_{jik} + J_{kij}}{3}.$$

Its norm satisfies

$$\|\operatorname{Sym} J\|^2 = \frac{\|J\|^2 + 2\langle J, J^{(12)} \rangle}{3}, \quad \langle S, \operatorname{Sym} J \rangle = \langle S, J \rangle.$$

Consequently the preceding integrand equals

$$2\|S + \operatorname{Sym} J\|^2 + \frac{1}{6}\|J - J^{(12)}\|^2 \geq 0. \quad (6.10)$$

This pointwise calculation proves (6.1); combining it with (6.3) proves (6.2). $\square$

For the soft slab potentials, H is the sum of a positive quadratic part and the positive semidefinite Hessians of the homogeneous summands. Lemma 6.1 will control each summand through the two terms on the right side of (6.2).

7 The variance inequality

We now apply the moment-coordinate estimates to the potential (2.1). In this section all expectations are with respect to its probability μ .

Proof of Proposition 2.1. The potential is smooth and even, and $D^2V \geq \varepsilon I$. Lemma 3.1 therefore supplies the moment coordinates, with $0 < \tau \leq (4/\varepsilon)I$. Put

$$H_i = D_x^2 \psi_i, \quad a_i = \mathbb{E} \psi_i.$$

Each H_i is positive semidefinite, and $a_i > 0$ since $p_i \neq 0$ and μ has positive density everywhere. Homogeneity gives

$$x \cdot \nabla \psi_i = q \psi_i, \quad H_i x = (q - 1) \nabla \psi_i. \quad (7.1)$$

The adjunction identity of Section 4 also gives

$$m_i := \mathbb{E} \operatorname{tr}(\tau H_i) = \mathbb{E} x \cdot \nabla \psi_i = q a_i. \quad (7.2)$$

To justify it for these noncompact functions, apply $\mathbb{E} x_j k = \mathbb{E} \tau_{jk} \partial_{x_k} k$ to $k = \eta_R \partial_{x_j} \psi_i$ and sum in j , where η_R is a smooth cutoff equal to one on B_R and zero outside B_{2R} . The extra term is bounded in absolute value by $CR^{-1} \mathbb{E}[|\nabla \psi_i| \mathbf{1}_{\{|x| \geq R\}}]$, which tends to zero. The other terms converge because τ is bounded and all polynomial moments are finite.

Take an even $f \in C_c^\infty(\mathbb{R}^n)$ and use

$$B = \tau D_x^2 f \tau, \quad W = \delta_\mu B, \quad u = \delta_\mu W = A(A - 1)f, \quad M = \tau^{-1}.$$

Applying (4.5) to $F = \psi_i$ gives, for each i ,

$$d_i := \mathbb{E} u \psi_i = \mathbb{E} W \cdot \nabla \psi_i = \mathbb{E} \operatorname{tr}(B H_i). \quad (7.3)$$

Two Cauchy–Schwarz estimates will give complementary bounds for these same numbers d_i .

First apply the Hilbert–Schmidt Cauchy–Schwarz inequality, including integration, to the matrix fields $H_i^{1/2} \tau^{1/2}$ and $H_i^{1/2} B \tau^{-1/2}$. Their pairing is $\operatorname{tr}(H_i B)$, so

$$d_i^2 \leq m_i \mathbb{E} \operatorname{tr}(M B H_i B). \quad (7.4)$$

This does not require B to be positive semidefinite. Second, (7.1)–(7.3) give

$$\mathbb{E} x^T H_i x = (q - 1) m_i, \quad \mathbb{E} x^T H_i W = (q - 1) d_i.$$

Cauchy–Schwarz for the positive semidefinite form $(X, Y) \mapsto \mathbb{E} X^T H_i Y$ therefore yields

$$\mathbb{E} W^T H_i W \geq (q - 1) \frac{d_i^2}{m_i}. \quad (7.5)$$

Since $H = D^2V = \varepsilon I + \sum_i H_i$, Lemma 6.1 and the weighted Bochner identity (6.3) imply

$$\begin{aligned} \mathbb{E} u^2 &\geq \mathbb{E} \operatorname{tr}(M B H B) + \mathbb{E} W^T H W \\ &\geq \sum_i \left(\mathbb{E} \operatorname{tr}(M B H_i B) + \mathbb{E} W^T H_i W \right) \geq \sum_i \frac{q d_i^2}{m_i} = \sum_i \frac{d_i^2}{a_i}. \end{aligned} \quad (7.6)$$

The discarded terms are nonnegative, since M is positive definite and B is symmetric. This is the required estimate on the dense class of double divergences.

By Lemma 5.1, such u are dense in the centered even subspace of $L^2(\mu)$. Each $\psi_i \in L^2(\mu)$, so all pairings $u \mapsto \mathbb{E}u\psi_i$ are continuous. Thus (7.6) holds for every centered even $u \in L^2(\mu)$. For $F = \sum_i b_i \psi_i$, choose $u = F - \mathbb{E}F$. Then

$$\|u\|_2^2 = \sum_i b_i d_i \leq \left(\sum_i b_i^2 a_i \right)^{1/2} \left(\sum_i d_i^2 / a_i \right)^{1/2} \leq \left(\sum_i b_i^2 a_i \right)^{1/2} \|u\|_2.$$

If $\|u\|_2 = 0$, the conclusion is immediate; otherwise division and squaring prove (2.2). This completes the analytic assertion and, by Section 2, Theorem 1.1. $\square$

8 Volume and measure consequences

With Theorem 1.1 proved, we derive the additive volume inequality stated in the introduction and then apply Saroglou's transfer theorem to even log-concave measures.

8.1 The symmetric L_p volume inequality

Proof of Corollary 1.2. At $\lambda = 0, 1$, the assertion follows from $\mathcal{W}[h_K] = K$ and $\mathcal{W}[h_L] = L$. Suppose $0 < \lambda < 1$. Set

$$a = |K|^{1/n} > 0, \quad b = |L|^{1/n} > 0, \quad K_0 = a^{-1}K, \quad L_0 = b^{-1}L,$$

so that $|K_0| = |L_0| = 1$. Define

$$c = ((1 - \lambda)a^p + \lambda b^p)^{1/p} > 0, \quad \theta = \frac{\lambda b^p}{c^p} \in (0, 1).$$

Then $1 - \theta = (1 - \lambda)a^p / c^p$. For every $u \in S^{n-1}$, the support identities $h_K = ah_{K_0}$ and $h_L = bh_{L_0}$, followed by the weighted arithmetic-geometric mean inequality, give

$$\begin{aligned} ((1 - \lambda)h_K(u)^p + \lambda h_L(u)^p)^{1/p} &= c((1 - \theta)h_{K_0}(u)^p + \theta h_{L_0}(u)^p)^{1/p} \\ &\geq ch_{K_0}(u)^{1-\theta} h_{L_0}(u)^\theta. \end{aligned}$$

All support values here are positive. The Wulff definition is monotone in its support bound and satisfies $\mathcal{W}[cf] = c\mathcal{W}[f]$ for $c > 0$. Hence

$$c\mathcal{W}[h_{K_0}^{1-\theta} h_{L_0}^\theta] \subset \mathcal{W}[(1 - \lambda)h_K^p + \lambda h_L^p]^{1/p}.$$

By Theorem 1.1, the unscaled body on the left has volume at least 1, because K_0, L_0 are origin-symmetric and have unit volume. The body on the right therefore has volume at least c^n . Raising this bound to the positive power p/n gives (1.2), since $c^p = (1 - \lambda)|K|^{p/n} + \lambda|L|^{p/n}$. $\square$

8.2 Even log-concave measures and the (B) -conjecture

We now pass from volume to integration against even log-concave measures. A nonnegative Radon measure μ on $\mathbb{R}^n$ is *log-concave* if

$$\mu((1 - \theta)A + \theta B) \geq \mu(A)^{1-\theta} \mu(B)^\theta \quad (0 < \theta < 1)$$

for all compact sets $A, B \subset \mathbb{R}^n$; it is *even* if $\mu(A) = \mu(-A)$ for every Borel set A . Radon measures are finite on compact sets, but their total mass need not be finite. We use the analogous multiplicative definition of log-concavity for nonnegative functions, so the identically zero function is included. Endpoint products in an interpolation inequality are understood as the corresponding endpoint values.

Saroglou's transfer theorem states that the logarithmic Brunn–Minkowski inequality for Lebesgue measure in a dimension d implies the same inequality in dimension d for every even log-concave density [29, Theorem 3.1]. The density may vanish and need not be normalized. This theorem is stated for densities with respect to d -dimensional Lebesgue measure. The reduction below makes explicit how it also gives the measure statement when the measure has lower-dimensional support.

Corollary 8.1 (Measure inequality and scalar (B) -conjecture). *Let $n \geq 1$, let μ be an even log-concave Radon measure on $\mathbb{R}^n$, and let $K, L \subset \mathbb{R}^n$ be origin-symmetric convex bodies. Then, for every $0 \leq \lambda \leq 1$,*

$$\mu(\mathcal{W}[h_K^{1-\lambda} h_L^\lambda]) \geq \mu(K)^{1-\lambda} \mu(L)^\lambda. \quad (8.1)$$

Consequently, for every origin-symmetric convex body $K \subset \mathbb{R}^n$, the function

$$\mathbb{R} \ni t \longmapsto \mu(e^t K) \quad (8.2)$$

is log-concave.

Proof. Fix $0 < \lambda < 1$ and write $D = \mathcal{W}[h_K^{1-\lambda} h_L^\lambda]$. The assertion is immediate when μ is the zero measure. Otherwise, choose a closed ball C centered at the origin large enough that $K \cup L \cup D \subset C$ and $0 < \mu(C) < \infty$. This is possible because the three bodies are compact and μ is a nonzero Radon measure. The probability measure

$$\nu(A) = \frac{\mu(A \cap C)}{\mu(C)}$$

is even and log-concave: for compact A, B , convexity of C gives

$$(1 - \theta)(A \cap C) + \theta(B \cap C) \subset ((1 - \theta)A + \theta B) \cap C,$$

and the defining inequality for μ applies. Since $K, L, D \subset C$, the common factor $\mu(C)$ cancels from (8.1). It therefore suffices to prove that inequality for ν .

Let E be the affine hull of $\text{supp } \nu$. Evenness gives $E = -E$, so E is a linear subspace. If $\dim E = 0$, then $\nu = \delta_0$; all three bodies contain 0, and equality holds. Suppose $d = \dim E \geq 1$ and identify E isometrically with $\mathbb{R}^d$. By inner regularity, the compact-set definition of log-concavity for ν is equivalent to the Borel-set formulation using inner measure. Borell's characterization [4, Theorem 1.1, with $s = 0$], applied to ν on E , represents it as an affine image of a log-concave density in a dimension $k \leq d$. Since its support spans E , the affine map has rank d . Thus $k = d$, the map is invertible, and ν has a log-concave density f with respect to Lebesgue measure on E . Evenness of ν gives $f(x) = f(-x)$ almost everywhere; replacing $f(x)$ by $\sqrt{f(x)f(-x)}$ gives an even log-concave representative of the same density.

Set $K_E = K \cap E$ and $L_E = L \cap E$. These are origin-symmetric convex bodies in E , since K and L contain neighborhoods of 0. Define their relative support functions for every $v \in E$ by

$$h_{K_E}^E(v) = \max_{y \in K_E} \langle y, v \rangle, \quad h_{L_E}^E(v) = \max_{y \in L_E} \langle y, v \rangle.$$

The relative Wulff body in E is

$$D_E = \bigcap_{v \in E} \left\{ x \in E : \langle x, v \rangle \leq h_{K_E}^E(v)^{1-\lambda} h_{L_E}^E(v)^\lambda \right\}.$$

Using all $v \in E$ rather than only unit vectors gives the same body by homogeneity. Let P_E be the orthogonal projection onto E . For $x \in D_E$ and $u \in S^{n-1}$,

$$\begin{aligned} \langle x, u \rangle &= \langle x, P_E u \rangle \\ &\leq h_{K_E}^E(P_E u)^{1-\lambda} h_{L_E}^E(P_E u)^\lambda \\ &\leq h_K(u)^{1-\lambda} h_L(u)^\lambda. \end{aligned}$$

Indeed, $h_{K_E}^E(P_E u) = \max_{y \in K \cap E} \langle y, u \rangle \leq h_K(u)$, and likewise for L ; if $P_E u = 0$, the middle product is 0. Thus

$$D_E \subset D \cap E. \tag{8.3}$$

Theorem 1.1 in dimension d , followed by Saroglou's transfer theorem [29, Theorem 3.1] in that same dimension, gives

$$\nu(D_E) \geq \nu(K_E)^{1-\lambda} \nu(L_E)^\lambda.$$

Since ν is supported on E , inclusion (8.3) yields

$$\nu(D) = \nu(D \cap E) \geq \nu(D_E) \geq \nu(K)^{1-\lambda} \nu(L)^\lambda,$$

as required. At $\lambda = 0, 1$, the Wulff bodies are K, L , respectively, and the endpoint convention gives equality.

Finally, for $s, t \in \mathbb{R}$, the support identity $h_{e^s K} = e^s h_K$ gives

$$\mathcal{W}[h_{e^s K}^{1-\lambda} h_{e^t K}^\lambda] = \mathcal{W}[e^{(1-\lambda)s + \lambda t} h_K] = e^{(1-\lambda)s + \lambda t} K.$$

Applying (8.1) to $e^s K$ and $e^t K$ proves

$$\mu(e^{(1-\lambda)s + \lambda t} K) \geq \mu(e^s K)^{1-\lambda} \mu(e^t K)^\lambda.$$

This is exactly the log-concavity in (8.2), and is the scalar-dilation argument of Saroglou [29, Corollary 3.2]. $\square$

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